Predicting MLB Hall of Famers

Emma Lynch, Katherine Snow, Margaret Potts
CSE 2600
December 9, 2025

Abstract

One of baseball's highest honors, induction into the MLB Hall of Fame, represents an excess of hard work, dedication, and talent in the sport. Accurately predicting which current players will get inducted is a problem that can be solved using data science techniques. In this project, multiple machine learning models were used to analyze and predict Hall of Fame induction using MLB career data from the Lahman Baseball Database. Logistic regression, random forest, and k-nearest neighbor models were created in the hopes to train and test the most accurate and reliable model in predicting the future induction of current players. Linear regression was used to identify which features of the data carry the greatest importance in the Hall of Fame selection process. Using the top features from that importance list, the random forest models demonstrated the best accuracy and reliability in predicting Hall of Fame status, while the logistic regression models performed with less accuracy, and the k-nearest neighbors model proved to be the worst at prediction.

Background

The Baseball Hall of Fame, located in Cooperstown, New York, serves as recognition of exceptional achievement in Major League Baseball. Each year, a committee votes on the eligibility of recently retired players, nominating several for a chance to be elected into the Hall of Fame. Induction is not based on any single metric, but rather on an overall evaluation of a player, considering many factors such as longevity of career and performance. The goal of this project is to identify the specific statistical factors most strongly associated with Hall of Fame selection and using this information to predict which active MLB players are most likely to be inducted into the Hall of Fame.

The Lahman Baseball Database, an open-source repository, contains detailed statistics from the 1871-2024 MLB regular seasons. The tables utilized to construct our models include Pitching, Batting, People, and HallOfFame. The records from the pitching and batting data sets contain stat lines from players' individual seasons. In order to build robust models, individual season statistics were aggregated into total career data for each player. Two separate models were created for the pitchers and the hitters, due to the completely different data variables that define success in these roles. For both batting and pitching, a random forest model was used to determine the relative importance of variables with respect to Hall of Fame induction and to

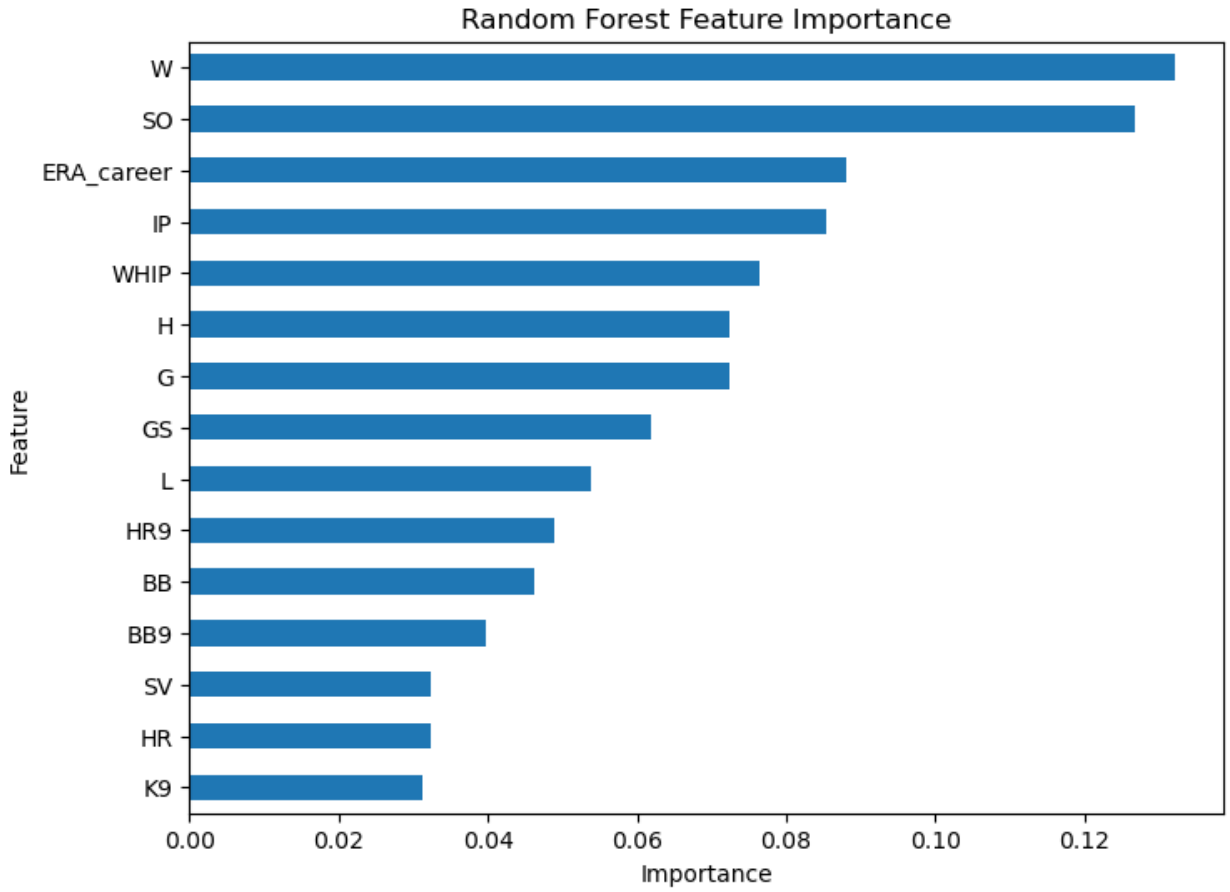
make predictions. Logistic regression and k-nearest neighbor models were also used to also make predictions. The reason multiple models were created was to compare accuracy and reliability between them in order to determine the most robust model. The models were trained on a portion of retired players' career data, whose Hall of Fame status is already known. Accuracy was found by testing our models on the remaining retired players. After creating a satisfactory model, the models were utilized to make predictions on which active MLB players would enter the Hall of Fame in the future.

Pitching

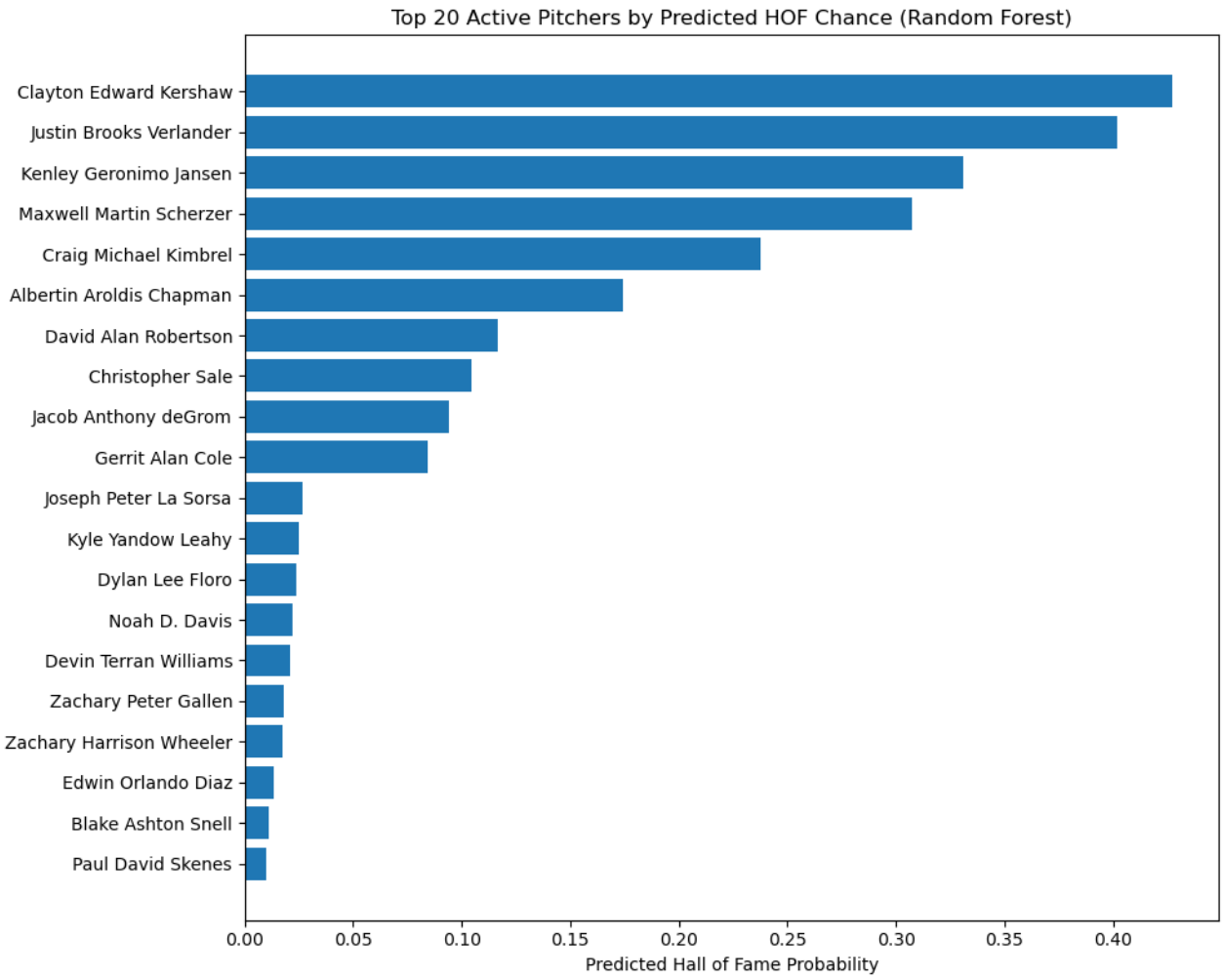
In order to properly evaluated player's pitching performance, the pitching CSV and hall of fame CSV were merged. Each player's statistics were aggregated into one career stat line. A binary feature was created to signify hall of fame inducted (1= inducted, 0 = not inducted). Lastly, players were sectioned into retired players and active players based on the last game they had played in the major leagues. To understand the relationship between a player's statistics, a random forest model was built in order to create a lists of variable importance. The features used to create the random forest model were innings pitched, wins, strikeouts, hits, games, walks, games started, losses, homeruns, and saves. Some features that were used and not found originally in the data set include career ERA (earned run average), homeruns per 9 innings, walks per 9 innings, strikeouts per 9 innings and WHIP (walks hits innings pitched). ERA is calculated by dividing a pitcher's earned runs by the number of innings they have pitched, multiplied by 9. WHIP is calculated by adding walks and hits together and divided by the number of innings pitched. ERA and WHIP were added into the model because they are commonly used stats to describe pitching performance. Homeruns per 9 innings, walks per 9 innings, and strikeouts per 9 innings were included because they normalize the stats to per 9 innings, so we can compare pitchers regardless of career length. The random forest model was trained on a portion of retired players and tested on the remaining portion of retired players. 75% of data was used to train and 25% of data was used to test. We needed to train and test on retired players because we know the hall of fame status of these players and can deduce the accuracy and performance of the model. Next, a logistic regression model was used to create another prediction. Logistic regression was chosen over linear regression because we are predicting a categorical variable – whether or not a player will enter the hall of fame, and we would need a probability to determine this.

The random forest model had an accuracy of 92%, with a false positive rate of 8% and a false negative rate of 0%. The area under the ROC curve was 0.98. The feature importance is as

follows:

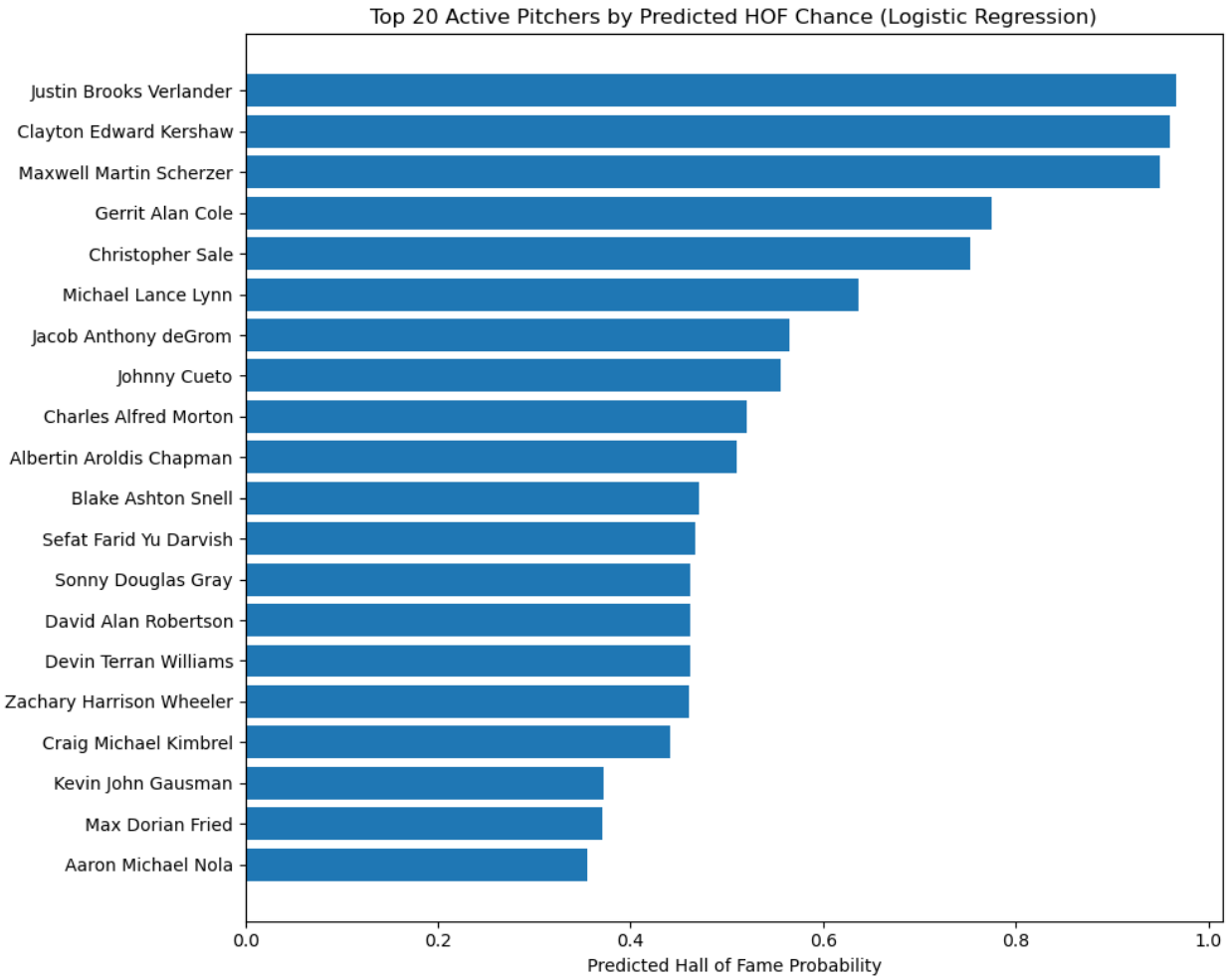


This list of feature importance suggests that a long career is what matters most in hall of fame consideration. Stats such as wins, strikeouts, and innings pitchers are all indicators of a long career. In addition to a long career, a player must have sustained high performances over each season, as suggested by the career ERA and WHIP stats also at the top of the list. In addition to creating the feature importance list, the random forest model was also used to make predictions on which actively playing baseball pitchers would enter the hall of fame.



In this model's predictions, the top players make sense: Clayton Kershaw, Justin Verlander, Kenley Jansen, and Max Scherzer are often considered the best pitchers in modern times. The model also predicts that players such as Gerrit Cole, Chris Sale, and Devin Williams have high chances of entering the hall of fame, but these players would need a couple more high-performance seasons to solidify their chances. Despite having only two seasons of major-league experience, Paul Skenes ranks 20th in the model's predictions. His inclusion reflects the model's capacity to capture standout rookie pitchers based on their strong statistical performance.

Using the top five features (W, SO, ERA_career, IP, WHIP), a logistic regression model was built and tested on retired players. The accuracy of the model was 83%, with a false positive rate of 17% and a false negative rate of 26%. The area under the ROC curve was 0.86. The model's top 20 current players that the model predicted to make the hall of fame are as follows:



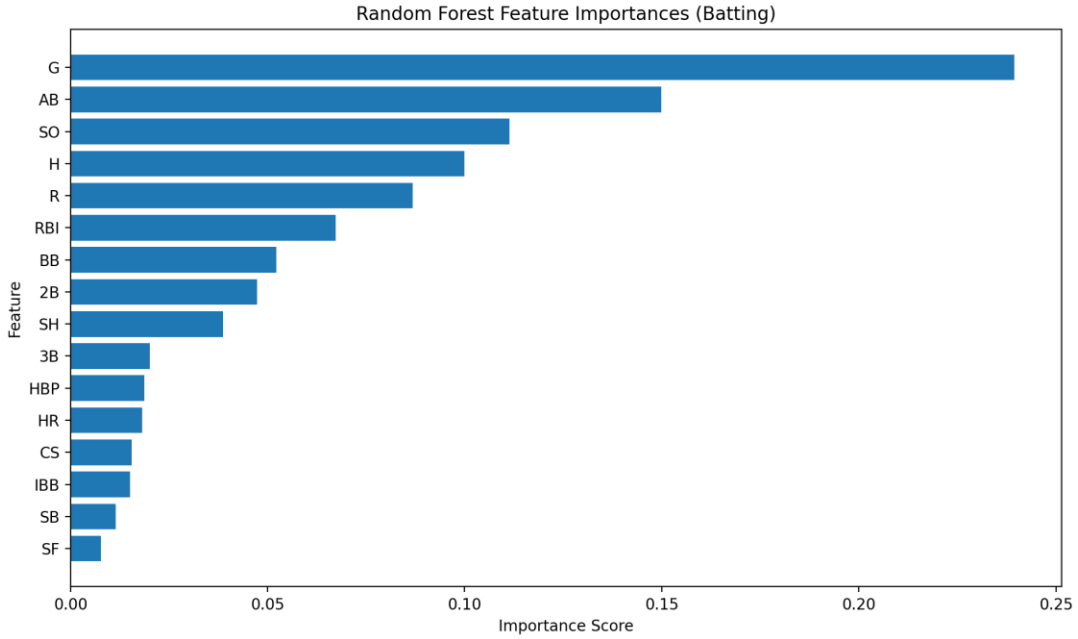
Multiple pitchers are featured on both lists. Justin Verlander, Clayton Kershaw, and Max Scherzer are still on top. One notable difference is that Kenley Jansen, who was ranked 2nd on the random forest model, is not in the top 20 of the logistic regression model. This is most likely because Jansen is closer, meaning that he normally only pitches in the last inning of the game. Because of this, he would not have a high frequency of innings pitched/strikeouts/wins in total. A reason for the drop in accuracy for the logistic regression model could be that since the logistic regression model is based on purely stats that suggest a long career, pitchers such as Jansen, who are still high-performance players but are closers/relievers will not show up in the predicted player list. Another key difference in the models is that numerical value associated with their probability. Because a random forest is built by averaging several trees together, and the vast majority of major league baseball players will not enter the hall of fame, the probabilities of the top players are lower.

Batting

The methodology used for batting performance data was similar to the methodology used for pitching performance data. Batting data was obtained from Batting.csv and merged with Hall of Fame induction information from HallOfFame.csv. A binary target variable HOF was created (1 = inducted, 0 = not inducted). The data was filtered to only get numeric, relevant statistics, and the resulting features used were Number of Games played, At-bats, Runs, Hits, 2nd Base Hits, 3rd Base Hits, Home Runs, RBIs, Stolen Bases, Caught Stealing, Base on Balls (Walks), Strikeouts, Intentional base on Balls, Hit By Pitch, Sacrifice Hits(Bunts), and Sacrifice Flies. The data was then aggregated to produce one row per player that contains the data across their entire career. This produced a structured dataset of career batting totals linked to a binary Hall of Fame outcome. No missing values were found in the batting dataset, so no imputation was required. We also made sure to balance class because we are using a highly imbalanced dataset, where we can expect the number of players who are predicted to be inducted to the Hall of Fame much smaller than the number of players who we can predict will not be inducted. Similar to the pitching models, we trained on retired players and tested on both retired and active players.

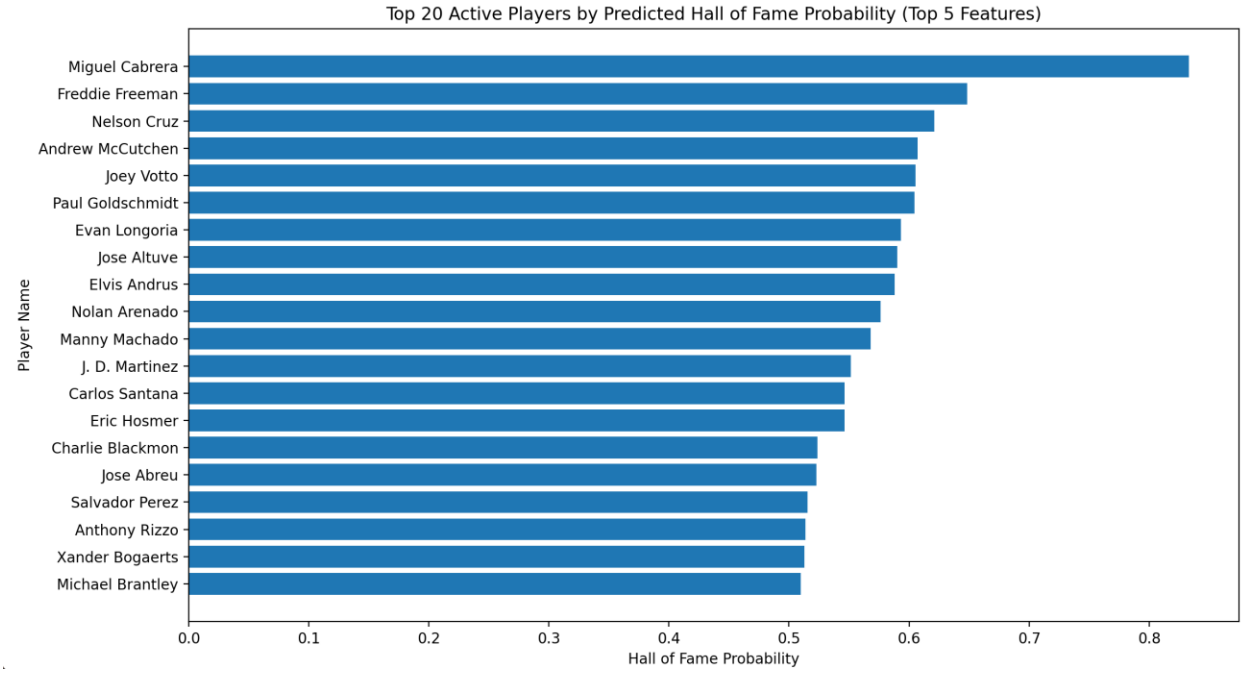
A series of three different models were used using batting performance data: Random Forest, Logistic Regression using the top 5 features found by Random Forest, and Logistic Regression using all features.

According to the Random Forest model, the top 5 features that are the most important in deciding Hall of Fame inductees are Number of Games played, At-Bats, Strikeouts, Hits, and Runs, as seen here:

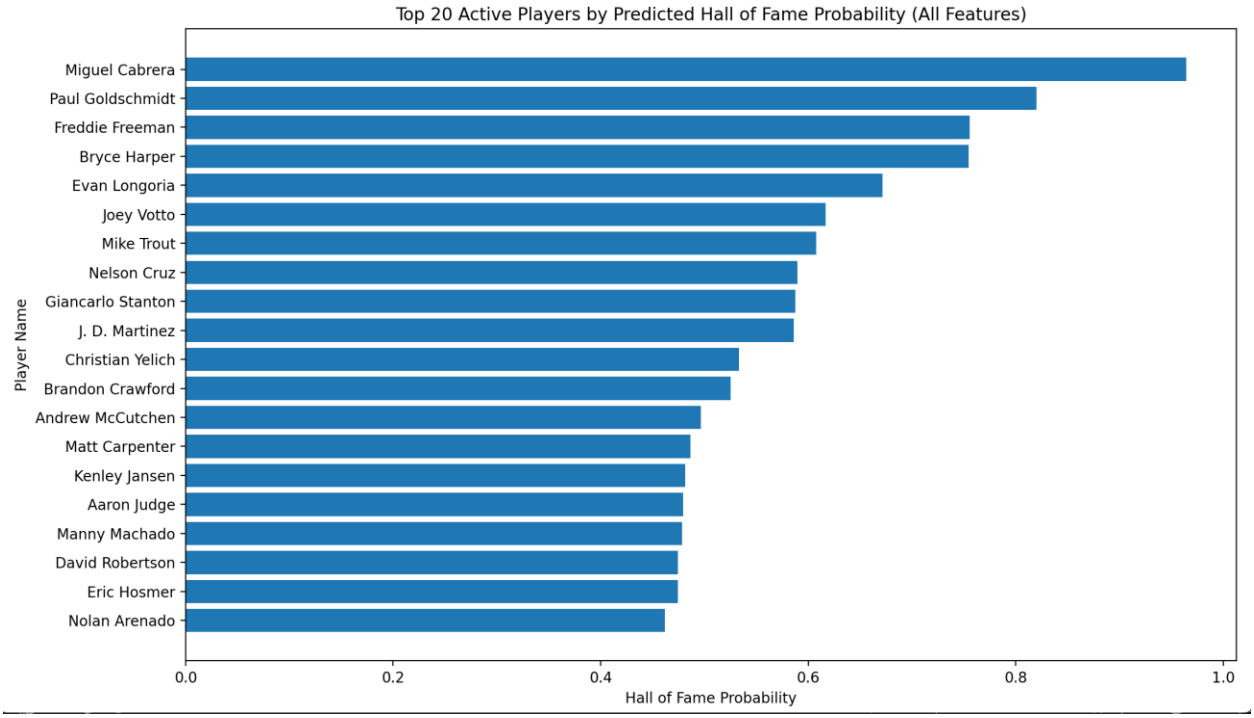


These results had 96% accuracy, a False Positive Rate of 3.9%, a False Negative Rate of 2.9%, and a ROC-AUC score of .98. The RF model was also used to determine the players most likely to be inducted, however the model's results showed accuracy of 97%, but a False Positive Rate of around 70%, meaning the model was massively overpredicting the amount of members that should be inducted. To fix this, we created an optimal threshold of 0.037 by subtracting the False Positive Rate from the True Positive Rate. With this threshold the RF model was only able to accurately predict the top 6 players rather than a top 20, meaning this model was not ideal.

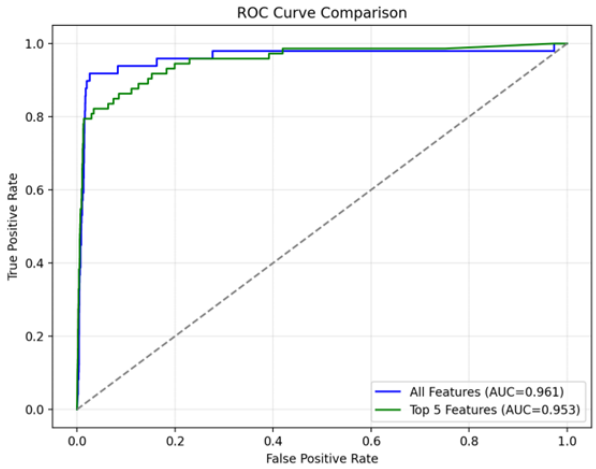
Next, we tested the Logistic Regression Model with only the top 5 features as determined by the RF model. The results had accuracy of 97%, a False Positive Rate of 2.4%, a False Negative Rate of 21%, and a ROC-AUC score of 0.953. We can see that Miguel Cabrera is the top pick to be in the Hall of Fame, while the following players show a rather similar likelihood of being inducted:



We also tested the Logistic Regression Model with every feature. These results had an accuracy of 97%, a False Positive Rate of 2.4%, a False Negative Rate of 21%, and a ROC-AUC score of 0.953. Again, Miguel Cabrera is the model's top pick to be inducted, with the following players having a more severe decrease in likelihood in this model than the Top 5 Features model:



Next, we need to find which of the logistic regression models is the most accurately predictive. First, we look at the ROC-AUC curve, where we can see that while both models perform well, the All Feature model performs slightly better:



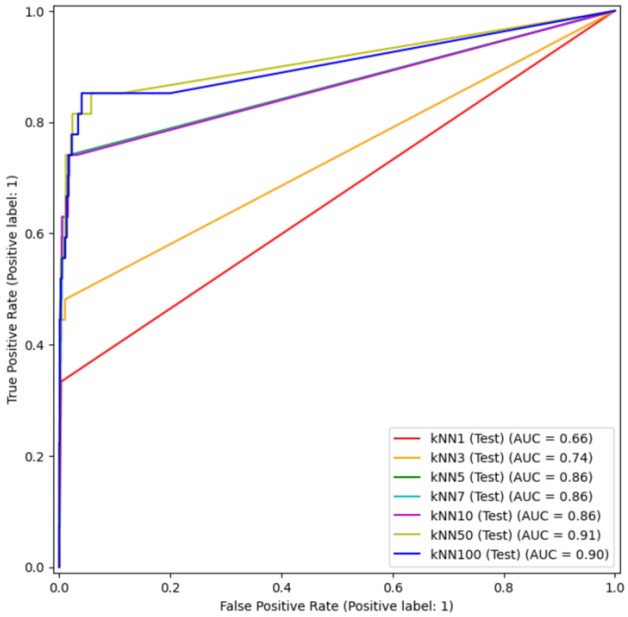
Cross-checking the names on the list with research on some of the players individually allowed for further analysis. Names like Miguel Cabrera, Freddie Freeman, and Joey Votto appeared on both lists, as they should have. However, the Top 5 features LR model is missing some top players that should definitely be on the list, like Mike Trout, Bryce Harper, and Aaron Judge. It also includes players like Elvis Andrus and Eric Hosmer who should definitely not be on the list. While the All Features LR model underestimates some players on the list that should be higher, they are at least included in the Top 20 players. The All Features model contains

players who make sense overall, while the Top 5 features model has some of these highly overestimated players.

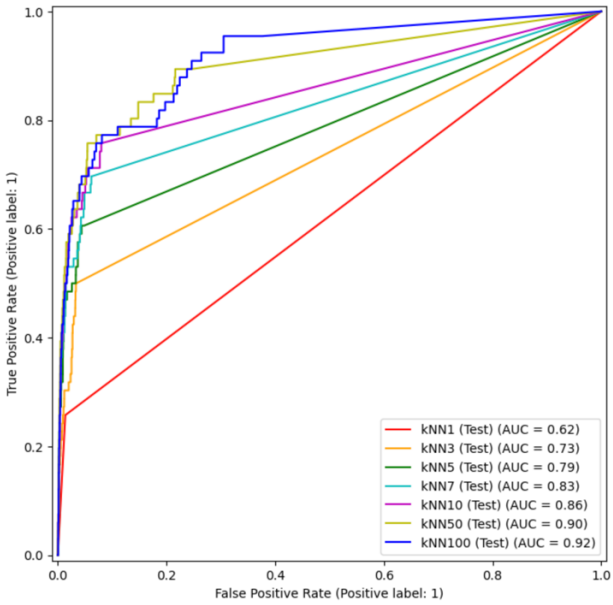
Overall, we conclude that the most efficient model for predicting Hall of Fame inductees is Logistic Regression using All Features. We know that Logistic Regression can perform better when using all features rather than just the top five because additional features may provide important context or correct for misleading correlations among top predictors. Removing features can force the model to rely too heavily on incomplete information, potentially reducing accuracy and predictive reliability. We found that the Top 5 predictors of Hall of Fame induction are Number of Games played, At-Bats, Strikeouts, Hits, and Runs. These things suggest that the most important factor in induction is career longevity. Pitching statistics tend to have more correlation with each other, as many pitching features are redundant. However, hitting statistics have less redundancy, meaning while we are able to determine a Top 5 most important features, it does not mean that only these Top 5 features allow us to accurately predict which players are most likely to be inducted with all the information necessary. Another limitation of this modeling approach is that only career totals were used as provided in the hitting dataset, which do not capture qualitative factors often considered in Hall of Fame voting, such as sportsmanship, or intangible peak career moments.

K-Nearest Neighbors

The same data preprocessing was used from the pitching and batting random forest models as the preprocessing for the pitching and batting k-nearest neighbors models. Features with a relative importance over 0.05 from the pitching and batting random forest importance lists were used to build the model. The models were also weighted by Euclidean distance, meaning that the neighbors that were the closest to the player being predicted were valued more in the decision process than the neighbors that were far away. The ROC curves for both the pitching and batting models were promising with the highest AUC for the pitching model being 0.91 (looking at 50 neighbors) and for the batting model the AUC was 0.92 (looking at 100 neighbors). Both of these AUC values would lead one to believe that both models have outstanding prediction capabilities, if that were the only metric seen.



Pitching K-Nearest Neighbors ROC Curve



Batting K-Nearest Neighbors ROC Curve

Despite the high AUC values, the k-nearest neighbors models had extremely low prediction power. The pitching model had a false positive rate of 0.0004 and a false negative rate of 0.778. The batting model had a false positive rate of 0 and a false negative rate of 1. Having false negative rates this high shows that both models are not predicting most of the actual Hall of Famers that made it into the hall of fame. The reason that the false positive rate is so high is that the proportion of players that get inducted into the hall of fame is extremely low, meaning most of the neighbors that the model is looking at did not get into the hall of fame. The percentage of

pitchers that have made it into the Hall of Fame is 1.01%, and the percentage of batters that have made it into the Hall of Fame is 1.26%. This also explains why the AUC values are so promising. Around 99 percent of players do not make it to the hall of fame, meaning that if one was to predict that all players did not get inducted, the accuracy would be around 99 percent.

```
Pitching Predictions                          Batting Predictions

Known Hall of Famers - knn50 wrighted model:  Known Hall of Famers - knn7 weighted model:
==============================================  ==============================================
Bob Gibson           | 100.0% | ✓ HOF         Hank Aaron           | 100.0% | ✓ HOF
Roy Halladay         | 100.0% | ✓ HOF         Derek Jeter          | 100.0% | ✓ HOF
Walter Johnson       | 100.0% | ✓ HOF         Chipper Jones        | 100.0% | ✓ HOF
Nolan Ryan           | 100.0% | ✓ HOF         Willie Mays          | 62.0% | ✓ HOF
Mike Mussina         | 32.7% | ✗ Not HOF      Babe Ruth            | 59.2% | ✓ HOF

Active Players - knn50 weighted model:        Active Players - knn7 weighted model:
==============================================  ==============================================
Max Scherzer         | 25.5% | ✗ Unlikely HOF  Miguel Cabrera       | 0.0% | ✗ Unlikely HOF
Craig Kimbrel        | 5.5% | ✗ Unlikely HOF   Nelson Cruz         | 0.0% | ✗ Unlikely HOF
Kenley Jansen        | 0.0% | ✗ Unlikely HOF   Freddie Freeman     | 0.0% | ✗ Unlikely HOF
Clayton Kershaw      | 0.0% | ✗ Unlikely HOF   Andrew McCutchen    | 0.0% | ✗ Unlikely HOF
Justin Verlander     | 0.0% | ✗ Unlikely HOF   Joey Votto          | 0.0% | ✗ Unlikely HOF
```

The k-nearest neighbors model performed poorly for Hall of Fame prediction, as shown in the sample results. While correctly identifying Hall of Famers, like Bob Gibson and Hank Aaron, it failed to recognize an established Hall of Famer, giving Mike Mussina only a 32.7% chance despite his actual induction. The model did not predict that any current star players would make it to the Hall of Fame, most of them having a 0% chance of induction. These unrealistic predictions demonstrate that the model cannot reliably distinguish between true Hall of Fame careers and ordinary ones in practice.

Reflection

The performances of the three classification models displayed different levels of predictive performance. The random forest models demonstrated extreme reliability, with an accuracy of 92% for pitchers and 96% for batters, and excellent AUC scores of 0.98 for both versions of the model. The results from the logistic regression models were just under that of the random forest. The best logistic models performed well with 83% accuracy and an AUC value of .86 for pitching and 97% accuracy and an AUC value of 0.95 for batting. All of these models significantly outperformed the k-nearest neighbors models in both accuracy and reliability. Despite having promising AUC scores of 0.91 and 0.92 for the pitching and batting models, the false negative rates were 0.778 and 1 respectively.

The superior performance of the random forest models is not surprising because they are able to handle imbalances datasets through the usage of ensemble learning, bootstrapped training samples, and averaging many decision trees. However, a major limitation across all models was

the severe class imbalance, where only about 1% of players achieve Hall of Fame status. This imbalance explained why the k-nearest neighbors model failed so dramatically.

Even though we had success with the random forest models, this project could be taken even further by making several adjustments. To reduce the impact of the overwhelming class imbalance, more sophisticated sampling techniques or cost-sensitive learning could be introduced to the models. This could be achieved by oversampling the known Hall of Famers, giving more weight to the Hall of Fame data, or implementing different ways to evaluate the models that take the imbalance into account.

If we were to repeat this project, we would not use k-nearest neighbors for prediction given the characteristics of the dataset. We would expand the amount of data used to include fielding and post-season statistics for a more complete player evaluation, and we would put more attention on which groups of features utilized could lead to the most accurate model. Additionally, we would incorporate qualitative features such as MVP awards or All-Star selections, which significantly influences real-world Hall of Fame voting.

References

https://sabr.org/lahman-database/